CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 39 (2012) 47–53

Contents lists available at SciVerse ScienceDirect

CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry

journal homepage: www.elsevier.com/locate/calphad

Thermodynamic properties of lead

J.W. Arblaster *

6 Brookside Close, Poolhouse Farm, Wombourne, Wolverhampton WV5 8JU, England, United Kingdom

ARTICLE INFO

Article history:
Received 13 June 2012
Received in revised form
30 August 2012
Accepted 30 August 2012
Available online 18 September 2012

Keywords:
Lead
Thermodynamic properties
Solid and liquid
Monatomic and diatomic gases
Enthalpies of sublimation
Vapour pressure curves

ABSTRACT

The thermodynamic properties of lead have been evaluated up to 2400 K. By discarding discrepant specific heat measurements included in previous reviews it has been possible to determine the specific heat curve of the solid phase to an accuracy exceeding 0.1% above 210 K. Selected values include an enthalpy of sublimation of 195.2 ± 0.4 kJ/mol for the monatomic gas phase at 298.15 K, a dissociation enthalpy D_0 of 82.1 ± 1.0 kJ/mol for the diatomic gas species at absolute zero and a derived equilibrium boiling point of 2020 K at one atmosphere pressure.

© 2012 Elsevier Ltd. All rights reserved.

1. Introduction

The superconducting transition temperature of 7.1997 ± 0.0025 K and the freezing point at 600.612 ± 0.001 K are secondary fixed points on ITS-90 [1]. Wherever possible values have been converted to the ITS-90 temperature scale [2,3] and to the currently accepted atomic weight of 207.2 ± 0.1 [4].

Previous reviews on lead were by Hultgren et al. [5], Gurvich et al. [6] and JANAF [7]. The values given for the condensed phases by CODATA [8] are the same as those of Gurvich et al. In all of these reviews specific heat measurements of Meads et al. [9] (14–300 K) were included in the room temperature region even though they were 1% higher than a combination of the measurements of Bronson and Wilson [10] and Leadbetter [11]. However Meads et al. also determined the specific heat of silver at the same time and in the room temperature region these measurements are also 1% higher than the values of Martin [12] which are considered to be the best measurements for this element. The measurements by Bronson and Wilson also determined on silver at the same time as the measurements on lead agree closely with those of Martin. It would appear therefore that the measurements of Meads et al. are subject to a systematic error and are rejected above 210 K but because of the poor quality of other data they are still considered to be the best measurements below this temperature.

2. Solid phase

For the superconducting state selected values given in Table 11 are based on a combination of the specific heat measurements of Phillips et al. [13] (0.3–0.8 K), as given by Phillips [14], and Neighbor et al. [15] (1.9–7.2 K).

In the normal state the following values have been obtained for the electronic coefficient (γ) and the limiting Debye temperature (Θ_D):- Table 1.

The selected values are the same as those selected by Phillips [14] whilst the limiting Debye temperature is in excellent agreement with the value 105.3 K determined by Waldorf and Alers [23] from elastic constant measurements.

Up to 8 K the selected values combine the above selected values of γ and Θ_D with the specific heat measurements of Van Der Hoeven and Keesom [19] (0.4–4.4 K) and Neighbor et al [15] (1.8–8.0 K). It is a requirement that at the superconducting transition temperature the entropy values for the two states are equal. The agreement to within 0.2 mJ/mol K as indicated in Table 11 is considered to be satisfactory.

From 8 to 90 K selected specific heat measurements of Meads et al [9] (14–300 K), Horowitz et al [16] (14–77 K) and Keesom and Kammerlingh Onnes [24] (14–81 K) were combined to give an overall accuracy of 1.4% although the measurements of Horowitz et al that are not included in the fit vary from 7% lower to 1% higher than the selected values whilst those of Keesom and Kamerlingh Onnes are from 9% lower to 2% higher. The three lowest data points of Meads et al averaged 9% lower and were also excluded.

From 90 to 210 K the measurements of Meads et al were selected but linearly adjusted in order to coincide with the selected value at 210 K. The accuracy of the fit in this region is 0.2%.

* Tel.: +44 01902 897394.
E-mail address: jwarblaster@yahoo.co.uk

0364-5916/$ - see front matter © 2012 Elsevier Ltd. All rights reserved.
http://dx.doi.org/10.1016/j.calphad.2012.08.004

Table 1
Low temperature specific heat coefficients.

Authors	Ref.	Temperature range (K)	γ (mJ/mol K^2)	Θ_D (K)
Horowitz et al.	[16]	1.4–4.1	3.13	96.3
Decker et al.	[17]	–	3.06	–
Keesom and Van Der Hoeven	[18]	0.4–4.4	2.98	106.7
Van Der Hoeven and Keesom	[19]	0.4–4.4	3.00	105.4
Clune and Green	[20]	1.0–2.3	3.04	105.7
Culbert et al.	[21]	0.5–1.7	3.00	104.1
Phillips et al.	[22]	0.3–0.8	3.00	105
Phillips et al.	[13]	0.3–0.8	2.99	104.9
	Selected		2.99	105.0

Above 210 K the specific heat measurements of Bronson and Wilson [10]. (203–393 K), Leadbetter [11] (301–587 K) and Malaspina et al [25] (305–455 K) were combined to give the following equation with an overall accuracy as a standard deviation of $\pm$ 0.026 J/mol K (0.09%):-

$$C_p^\circ\,(J/\text{mol K}) = 24.0968 + 8.78751 \times 10^{-3}T - 6.88913 \times 10^{-6}T^2 + 1.46403 \times 10^{-8}T^3 \quad (1)$$

The specific heat measurements of Meads et al. average 1% higher than the selected values above 210 K. Selected thermodynamic values between 8 and 293.15 K are given in Table 12 whilst values for the solid above 298.15 K are given in Table 13.

3. Liquid phase

Stølen and Grønvold [26] reviewed 41 determinations of the enthalpy of fusion and arrived at a weighted average of 4782 $\pm$ 11 J/mol which is accepted and this fixes the value for the liquid enthalpy at the melting point as $H_T^\circ - H_{298.15}^\circ = 13275 \pm 13$ J/mol. Grønvold [27] determined the specific heat at the melting point and 650 K and since these values confirmed those derived from the enthalpy measurements of Douglas and Dever [28] (623–1169 K) then these values were accepted. They showed a continuous decrease in derived specific heat with increasing temperature but from a comparison with tin it is expected that there would be a minimum. Gurvich et al [6] arbitrary selected a minimum of 1700 K whilst JANAF [7] selected 1400 K. However from the generation of a theoretical specific heat curve based on a combination of the method of pseudopotentials in the electron theory of metals and perturbation theory in statistical physics Onistchenko et al [29] arrived at a minimum at 1500 K which is selected. Therefore after correction to ITS-90 and from a 273.15 K base to a 298.15 K base the enthalpy measurements of Douglas and Dever were combined with the fixed value of $dC_p^\circ/dT = 0$ at 1500 K and after correction to the selected value of the enthalpy at the freezing point the following equation is derived with an overall accuracy of $\pm$ 23 J/mol (0.09%):-

$$H_T^\circ - H_{298.15}^\circ(J/\text{mol}) = 34.7186T - 4.06055 \times 10^{-3}T^2 + 9.02345 \times 10^{-7}T^3 - 6307.67 \quad (600.612 - 1500\ K) \quad (2)$$

With the correction to the selected value of the enthalpy at the freezing point the actual measurements of Douglas and Dever average 0.3% low. There are no satisfactory specific heat or enthalpy measurements above 1500 K and therefore the theoretical approach of Onistchenko et al is considered which suggests graphically that the difference between the specific heats of the liquid at the minimum at 1500 K and 2000 K is 8 J/kg K (1.658 J/mol K). Using these two values of specific heat and the fixed value $dC_p^\circ/dT = 0$ at 1500 K the derived equivalent enthalpy equation becomes

$$H_T^\circ - H^\circ{}_{298.15}(J/\text{mol}) = 43.5462T - 9.94560 \times 10^{-3}T^2 + 2.2101333 \times 10^{-6}T^3 - 10721.49 \quad (1500 - 2400\ K) \quad (3)$$

Values for the thermodynamic properties of the liquid are given in Table 13.

4. Monatomic gas

Values as given in Table 14 are based on one bar standard state pressure and are calculated from the nineteen energy levels below 50,000 cm^{-1} given by Wood and Andrew [30]. Thermodynamic properties were calculated using the method of Kolsky et al [31] and the 2010 Fundamental Constants [32].

5. Diatomic gas

Thermodynamic properties were calculated by Rand [33] from five levels which include the three levels X (0_g^+), F (0_u^+) and C (0_u^+) for which full spectroscopic constants are available [34–36] and the two low lying levels $X_2 1_g$ and $A_2 1_u$ for which only a limited number of spectroscopic constants were given in [37] with the remainder being estimated. The existence of a further 31 levels has been predicted by Das et al [38]. Derived thermodynamic properties are given in Table 15.

6. Enthalpy of sublimation of the monatomic gas

For values given in the form of Clausius–Clapeyron equation a "pseudo" Third Law value was calculated by evaluating the enthalpy of sublimation at the temperature extremes and averaging. The values obtained give a reasonable estimate compared to the values which would have been obtained if all of the data had been available. Because of a lack of detail as to what temperature scales were being used then no attempt was made to correct values to ITS-90 from what would be contemporary scales. Only high temperature boiling point determinations were corrected for the presence of the dimer. Vapour pressure measurements for the liquid given by Alcock and Grieveson [39] were not included since no temperature values were given.

The selected value was based on the last ten sets of measurements. In consideration of other reviews, the values of Harteck [46] and Schins et al [59] selected by Gurvich et al [6] were not selected in this evaluation because of a poor agreement between Second and Third Law values and similarly the values of Andon et al [60] selected by CODATA [8] were rejected for the same reason. The directly determined boiling point measurement of Fischer [49] as selected by Gurvich et al was assumed to be superseded by the later measurement of both Kirshenbaum and Cahill [70,71] and Bohdansky and Schins [73,74] (Table 2).

7. Dissociation enthalpies of the diatomic gas

Values are given for the Second Law and Third Law derived enthalpy of dissociation D_0 at absolute zero. The ionisation ratio I_{Pb2}^+/I_{Pb1}^+ given by Honig [80,81] was corrected to a Pb_2/Pb_1 pressure ratio assuming an ionisation cross section ratio $\sigma_{Pb2}/\sigma_{Pb1}$ of 1.5 (Table 3).

The accuracy assigned to the selected value takes in to account the accuracy assigned to the enthalpy of sublimation of the monatomic gas which was required for the calculations.

Table 2
Enthalpy of sublimation of the monatomic gas at 298.15 K.

Authors	Ref.	Method	Range (K)	ΔH (II) (kJ/mol)	ΔH (III) (kJ/mol)	Notes
Greenwood	[40]	BP	1798	–	175.5	a
Greenwood	[41]	BP	1583–2373	197 ± 8	178.8 ± 1.4	a
Von Wartenburg	[42]	BP	1080–1451	201 ± 1	192.0 ± 0.8	
Ruff and Bergdahl	[43]	BP	1548–1828	214 ± 7	180.2 ± 0.8	a
Ingold	[44]	BP	1195–1611	210 ± 1	190.5 ± 0.3	
Egerton	[45]	Eff	800–1046	204 ± 3	197.7 ± 0.2	
Harteck	[46]	KE	1353–1477	208 ± 5	194.2 ± 0.3	
Von Leitgebel	[47]	BP	2013	–	194.6	a
Brüll	[48]	BP	1883	–	183.0	a
Fischer	[49]	BP	2028	–	196.0	a
Baur and Brunner	[50]	BP	1329–1673	203 ± 1	193.5 ± 0.5	
Granovskaya and Lyubimov	[51]	KE	1200–1490	217 ± 9	196.1 ± 1.3	
Aldred and Pratt	[52]	TE	880–1050	217	196.0 ± 1.9	b
Esyutin and Onaev	[53]	Evap	1173–1473	193 ± 1	195.0 ± 0.1	
Dennison	[54]	Eff	845–1025	199 ± 1	193.5 ± 0.1	
Goldfinger and Jeunehomme	[55]	KE	913–1032	199 ± 5	195.8 ± 0.2	
Sieben and Schmahl	[56]	Trans	1139–1350	190 ± 3	195.9 ± 0.2	
Rice and Ragone	[57]	OA	–	–	197.7 ± 3.6	c
Khobdabergenov et al	[58]	Flow	1273–1473	202	191.9 ± 0.7	b
Schins et al	[59]	BP	1628–2177	208 ± 1	196.0 ± 0.2	a
Andon et al	[60]	KE	888–1060	171 ± 10	195.1 ± 0.7	
Desideri and Piacente	[61]	TE	1020–1200	–	194.6 ± 0.4	c
Kopach and Lukashenko	[62]	Eff	1040–1225	195 ± 2	195.8 ± 0.1	
Muradova	[63]	AA	769–896	175 ± 2	198.9 ± 0.3	b,d
Landsberg et al	[64]	KE	721–1128	205 ± 10	195.4 ± 1.3	b,e
		TE	867–1128	200 ± 22	193.1 ± 1.9	
Katsov et al	[65]	AA	1030–1184	195	–	
Vaisburd et al	[66]	Eff	773–1473	199 ± 4	197.6 ± 0.6	
Botor et al	[67]	TE	922–1073	198 ± 1	197.1 ± 0.1	
		KE	922–1073	200 ± 1	198.5 ± 0.1	
		DTE	990–1214	218 ± 5	199.3 ± 0.3	
Rodebush and Dixon	[68]	Static	1391–1508	200 ± 1	195.6 ± 0.1	
Rodebush and Dixon	[69]	BP	1433–1597	205 ± 7	195.3 ± 0.2	
Kirshenbaum and Cahill	[70,71]	BP	2022	–	195.4	a
Kim and Cosgarea	[72]	TE	877–975	197 ± 1	195.0 ± 0.1	
Bohdansky and Schins	[73,74]	BP	1628–2177	206 ± 2	195.2 ± 0.5	a
Hawkins and Hultgren	[75]	TE	950–1125	198 ± 1	195.2 ± 0.1	
Chang et al	[76]	TE	961–1078	199 ± 2	195.2 ± 0.1	f
Shiu and Munir	[77]	KE	950–1112	196 ± 1	195.0 ± 0.1	
Prasad et al	[78]	Trans	1174–1301	194 ± 1	195.0 ± 0.1	
Zajaczkowski and Botor	[79]	KE	907–1025	195 ± 1	195.4 ± 0.1	g
	Selected				**195.2 ± 0.4**	

Notes: (a) Corrected for the presence of the dimer; (b) given only in the form of Clausius–Clapeyron equation; (c) value given only at 298.15 K; (d) average of four runs. Two further runs 919–1110 K and 963–1128 K giving Third Law values of 190.6 ± 0.7 kJ/mol and 186.0 ± 0.7 kJ/mol respectively were not included in the average; (e) average of three runs. Two further runs 867–1021 K and 963–1128 K giving Third Law values of 185.9 ± 2.3 kJ/mol and 169.0 ± 2.2 kJ/mol respectively were not included in the average; (f) one data point at 1057.5 K rejected; (g) a second run 976–1132 K was rejected by the authors.
Methods: *BP*: Boiling point; Eff: Effusion; *KE*: Knudsen effusion; *OA*: Optical absorption; *TE*: Torsion effusion; *AA*: Atomic absorption; *DTE*: Differential torsion effusion; *Trans*: Transport; *Evap*: Evaporation.

Table 3
Dissociation enthalpy of the diatomic gas at 0 K.

Authors	Ref.	Method	Range (K)	D_0 (II) (kJ/mol)	D_0 (III) (kJ/mol)
Honig	[80,81]	MS	800	–	96
Gingerich et al	[82]	KEMS	900–1117	83.5 ± 2	82.1 ± 0.1
Nabot and Chatillon	[83]	KEMS	1400–1725	–	110 ± 11
	Selected				**82.1 ± 1.0**

Methods: *MS*: Mass spectrometry; *KEMS*: Knudsen effusion mass spectrometry.

8. Vapour pressure

Values for solid lead and the Pb_1 gas were evaluated at 10 K intervals from 450 K to 600 K and the freezing point. Values for liquid lead, Pb_1 and Pb_2 were calculated at 50 K intervals from 650 K to 2100 K and the freezing point. Because of the complex behaviour of the specific heat curve of the liquid it was found that best agreement with thermodynamically derived vapour pressures could be achieved if two equations were used in each case to represent the data (Table 4).

$$\ln(p,\text{bar}) = A + B\ln(T) + C/T + D\,T + E\,T^2$$

9. A comparison of selected values at 298.15 K

Values for specific heat and enthalpy given in the present review differ notably from those selected previously because in this case the discrepant values of Meads, Forsythe and Giauque [9] were not included above 210 K (Table 5).

10. A summary of thermodynamic equations

See Tables 6–8.

Table 4
Vapour pressure equations.

Phase	Range (K)	A	B	C	D	E
Solid + Pb_1	450–600.612	17.61639	−0.6499917	−23671.77	1.95686×10^{-4}	-3.23478×10^{-7}
Liquid + Pb_1	600.612–1350	23.04539	−1.585081	−23466.85	3.73167×10^{-4}	-2.99718×10^{-8}
	1350–2100	26.11518	−2.018750	−23734.11	5.74303×10^{-4}	3.95906×10^{-8}
Liquid + Pb_2	600.612–1350	39.89279	−3.544486	−36951.54	6.35211×10^{-4}	-1.60450×10^{-8}
	1350–2100	62.37680	−6.853330	−38580.04	2.88738×10^{-3}	-2.73095×10^{-7}
Liquid + Pb_1 + Pb_2	600.612–1350	22.33188	−1.469254	−23436.83	2.25293×10^{-4}	1.33243×10^{-9}
	1350–2100	28.44841	−2.334631	−23966.53	6.34222×10^{-4}	-1.98988×10^{-8}

Table 5
A comparison of specific heat, enthalpy and entropy at 298.15 K.

Authors	Ref.	C_p J/mol K	$H^\circ_{298.15} - H^\circ_0$ J/mol	$S^\circ_{298.15}$ J/mol K
Hultgren et al.	[5]	26.65	6874	65.06
Gurvich et al.	[6]	26.65	6870	64.80
JANAF	[7]	26.84	6878	64.79
This work	–	26.49	6846	64.87

Table 6
Representative equations above 298.15 K.

Solid: 298.15–600.612 K

$$C^\circ_P(\mathrm{J/mol\ K}) = 24.0968 + 8.78751 \times 10^{-3}T - 6.88913 \times 10^{-6}T^2 + 1.46403 \times 10^{-8}T^3 \quad (4)$$

$$H^\circ_T - H^\circ_{298.15}(\mathrm{J/mol}) = 24.0968\,T + 4.393755 \times 10^{-3}T^2 - 2.296377 \times 10^{-6}T^3 + 3.660075 \times 10^{-9}\,T^4 - 7543.10 \quad (5)$$

$$S^\circ_T(\mathrm{J/mol\ K}) = 24.0968\ln(T) + 8.78751 \times 10^{-3}T - 3.444565 \times 10^{-6}T^2 + 4.88010 \times 10^{-8}T^3 - 74.8633 \quad (6)$$

Liquid: 600.612–1500 K

$$C^\circ_P\ (\mathrm{J/mol\ K}) = 34.7186 - 8.12110 \times 10^{-3}T + 2.707035 \times 10^{-6}T^2 \quad (7)$$

$$H^\circ_T - H^\circ_{298.15}\ (\mathrm{J/mol}) = 34.7186T - 4.06055 \times 10^{-3}T^2 + 9.02345 \times 10^{-7}T^3 - 6307.67 \quad (8)$$

$$S^\circ_T(\mathrm{J/mol\ K}) = 34.7186\ln(T) - 8.12110 \times 10^{-3}T + 1.3535175 \times 10^{-6}T^2 - 125.3771 \quad (9)$$

Liquid: 1500–2400 K

$$C^\circ_P\ (\mathrm{J/mol\ K}) = 43.5462 - 1.98912 \times 10^{-2}T + 6.63040 \times 10^{-6}T^2 \quad (10)$$

$$H^\circ_T - H^\circ_{298.15}\ (\mathrm{J/mol}) = 43.5462T - 9.94560 \times 10^{-3}T^2 + 2.2101333 \times 10^{-6}T^3 - 10721.49 \quad (11)$$

$$S^\circ_T\ (\mathrm{J/mol\ K}) = 43.5462\ln(T) - 1.98912 \times 10^{-3}T + 3.31520 \times 10^{-6}T^2 - 176.6940 \quad (12)$$

Table 7
Free energy equations above 298.15 K.

Solid: 298.15–600.612 K

$$G^\circ_T - H^\circ_{298.15}\ (\mathrm{J/mol}) = 98.9601\,T - 4.393755 \times 10^{-3}T^2 + 1.148188 \times 10^{-6}T^3 - 1.220025 \times 10^{-9}T^4 - 24.0968\,T\ln(T) - 7543.10 \quad (13)$$

Liquid: 600.612–1500 K

$$G^\circ_T - H^\circ_{298.15}\ (\mathrm{J/mol}) = 160.0957T + 4.06055 \times 10^{-3}T^2 - 4.511725 \times 10^{-7}T^3 - 34.7186\,T\ln(T) - 6307.67 \quad (14)$$

Liquid: 1500–2400 K

$$G^\circ_T - H^\circ_{298.15}\ (\mathrm{J/mol}) = 220.2402T + 9.94560 \times 10^{-3}T^2 - 1.1050667 \times 10^{-6}T^3 - 43.5462\,T\ln(T) - 10721.49 \quad (15)$$

Table 8
Transition values involved with the free energy equations.

Transition	T (K)	ΔH° (J/mol)	ΔS° (J/mol K)
Fusion	600.612	4782.000	7.9619

Table 9
Specific heat measurements

Authors	Ref.	Range (K)	Deviations
Superconducting state			
Horowitz et al	[16]	1.8–3.7	Scatter 1% high to 49% high
Keesom and Van Der Hoeven	[18]	0.3–4.4	Bias 4% high
Keesom and Van Der Hoeven	[19]	0.4–4.5	Trend 12% high to 2% high
Keesom and Andrews	[84]	2.2–6.3	Above 4 K trend 400% high to 49% high
Keesom and Van Den Ende	[85]	2.2–6.2	Above 3 K trend 136% high to 16% high
Bonnerot	[86]	3–7 4.7–4.9	Bias 5% high (first run) Bias 11% high (second run)
Normal state			
Keesom and Andrews	[84]	2.2–16.4	Bias 11% low from 12 to 16 K
Bonnerot	[86]	3–40	Trend 16% low to 4% high
Drucker	[87]	14–78	Above 18 K scatter 1% low to 2% high
Eucken and Schwers	[88]	16–276	Bias 2% high above 22 K
Nernst	[89]	23–273	Scatter 6% low to 7% high
Barschall	[90]	90	6% high
Griffiths and Griffiths	[91]	119–371	Bias 0.3% low
Klinkhardt	[92]	323–600	Trend 1% high to 4% high
Brandt and Schulz	[93]	273–473	Trend 2% high to 3% low
Bartenev	[94]	273–473	Trend 2% high to 5% high
Bartenev	[95]	323–573	Scatter 2% low to 19% high
Pochapsky	[96]	273–588	Trend 1% low to 3% high
Kogan and Lyusternik	[97]	293–593	Trend to 3% low
Cordoba and Brooks	[98]	300–600	Bias 0.8% low
Liquid state			
Klinkhardt	[92]	600–773	Trend 5% high to 9% high
Brandt and Schulz	[93]	600–800	Bias 1% low
Bartenev	[95]	623–673	Bias 14% high
Kogan and Lyusternik	[97]	643–793	Bias 41% high
Cordoba and Brooks	[98]	601–850	Trend 2% low to 9% low
Dixon and Rodebush	[99]	627–732	Bias 4% low
Banchilla and Filippov	[100]	1150–2050	Trend 1% low to 10% low

Table 10
Enthalpy measurements.

Solid state			
Douglas and Dever	[28]	373–573	Bias 0.2% low
Stücker	[101]	373–573	Trend 1% low to 2% high
Magnus	[102]	373–529	Bias 0.2% high
Schübel	[103]	373–575	Scatter 1% low to 0.8% high
Wüst, Meuthen and Durrer	[104]	373–593	Scatter but average bias 3% high
Iitaka	[105]	432–577	Bias 0.5% low
Umino	[106]	323–573	Trend 18% low to 4% low
Awberry and Griffiths	[107]	373–573	Trend 0.7% low to 10% low
Sommelet	[108]	571–589	Bias 1.6% low
Glaser	[109]	471	0.8% high
Schimpff	[110]	373	0.4% high

Table 10 (*continued*)

Solid state			
Liquid state			
Wüst et al	[104]	613–1273	Bias 3% low
Iitaka	[105]	622–914	Scatter 1% low to 1% high
Umino	[106]	623–1073	Bias 3% low
Awberry and Griffiths	[107]	627–759	Trend to 8% low
Sommelet	[108]	690–1256	Trend from 1% low to agreeing with fit
Glaser	[109]	633–671	Trend 7% low to 0.3% high
Mishra and Kumar	[111]	625–1175	Trend 1% high to 0.4% low
Pottlacher and Jager	[112]	2000–5000	Bias 10% high over range 2000–2500 K

Not included: Kramer and Nölting [113] 180–600 K data shown only graphically.

Table 11
A comparison of thermodynamic properties below the superconducting temperature.

T (K)	**Superconducting state**			**Normal state**		
	C_p° (mJ/mol K)	$H_T^\circ - H_0^\circ$ (mJ/mol)	S_T° (mJ/mol K)	C_p° (mJ/mol K)	$H_T^\circ - H_0^\circ$ (mJ/mol)	S_T° (mJ/mol K)
0.5	0.211	0.0263	0.0702	1.705	0.400	1.565
1.0	1.684	0.421	0.561	4.681	1.917	3.552
1.5	5.706	2.139	1.903	10.25	5.512	6.393
2.0	13.92	6.813	4.538	19.81	12.83	10.54
2.5	28.73	17.18	9.104	35.14	26.28	16.48
3.0	52.03	36.95	16.25	58.24	49.24	24.79
3.5	87.36	71.21	26.74	92.29	86.33	36.15
4.0	139.3	127.1	41.59	141.7	144.1	51.51
4.5	213.0	214.2	62.01	211.7	231.5	72.02
5.0	313.6	344.6	89.41	307.0	360.0	99.01
5.5	445.3	532.9	125.2	431.4	543.7	133.7
6.0	611.6	795.7	170.8	587.4	796.7	177.8
6.5	813.3	1150	227.5	766.4	1136	232.1
7.0	1049	1615	296.2	998.0	1579	297.6
7.1997	1154	1835	327.2	1095	1787	327.0

Table 12
Low temperature thermodynamic data.

T (K)	C_p° (J/mol K)	$H_T^\circ - H_0^\circ$ (J/mol)	S_T° (J/mol K)	$-(G_T^\circ - H_0^\circ)/T$ (J/mol K)
8	1.527	2.879	0.464	0.104
10	3.390	7.817	1.008	0.226
15	7.517	35.39	3.187	0.828
20	10.956	81.84	5.836	1.744
25	13.790	143.9	8.595	2.838
30	16.099	218.9	11.321	4.026
35	17.955	304.2	13.948	5.257
40	19.426	397.8	16.445	6.501
45	20.576	497.9	18.803	7.738
50	21.459	603.1	21.018	8.956
60	22.630	824.1	25.044	11.309
70	23.285	1054	28.587	13.530
80	23.687	1289	31.724	15.612
90	24.009	1527	34.533	17.561
100	24.302	1769	37.078	19.387
110	24.554	2013	39.406	21.103
120	24.770	2260	41.552	22.719
130	24.954	2509	43.542	24.245
140	25.112	2759	45.398	25.690
150	25.247	3011	47.135	27.063
160	25.364	3264	48.768	28.369
170	25.465	3518	50.309	29.615
180	25.555	3773	51.767	30.805
190	25.634	4029	53.151	31.945
200	25.707	4286	54.468	33.039
210	25.773	4543	55.724	34.089

Table 12 (*continued*)

T (K)	C_p° (J/mol K)	$H_T^\circ - H_0^\circ$ (J/mol)	S_T° (J/mol K)	$-(G_T^\circ - H_0^\circ)/T$ (J/mol K)
220	25.853	4801	56.924	35.100
230	25.932	5060	58.075	36.074
240	26.011	5320	59.181	37.014
250	26.092	5581	60.244	37.922
260	26.173	5842	61.269	38.800
270	26.255	6104	62.258	39.651
280	26.339	6367	63.215	40.476
290	26.423	6431	64.140	41.276
298.15	26.492	6846	64.874	41.911

Table 13
High temperature thermodynamic data.

T (K)	C_p° (J/mol K)	$H_T^\circ - H_{298.15}^\circ$ (J/mol)	S_T° (J/mol K)	$-(G_T^\circ - H_{298.15}^\circ)/T$ (J/mol K)
Solid				
298.15	26.492	0	64.874	64.874
300	26.508	49	65.038	64.874
350	26.956	1385	69.157	65.199
400	27.447	2745	72.788	65.925
450	27.990	4131	76.052	66.872
500	28.598	5546	79.032	67.941
550	29.282	6992	81.789	69.076
600	30.052	8475	84.369	70.244
600.612	30.062	8493	84.399	70.258
Liquid				
600.612	30.817	13,275	92.361	70.258
700	30.360	16,315	97.046	73.738
800	29.954	19,331	101.073	76.910
900	29.602	22,308	104.580	79.793
1000	29.305	25,253	107.683	82.430
1100	29.061	28,171	110.464	84.854
1200	28.871	31,067	112.984	87.095
1300	28.736	33,947	115.289	89.177
1400	28.655	36,816	117.416	91.119
1500	28.628	39,679	119.391	92.938
1600	28.694	42,544	121.240	94.650
1700	28.893	45,423	122.985	96.266
1800	29.225	48,327	124.645	97.797
1900	29.689	51,272	126.237	99.252
2000	30.286	54,270	127.775	100.640
2100	31.015	57,333	129.270	101.968
2200	31.877	60,477	130.732	103.242
2300	32.871	63,713	132.170	104.469
2400	33.998	67,056	133.593	105.653

Table 14
Thermodynamic properties of the monatomic gas.

T (K)	C_p° (J/mol K)	$H_T^\circ - H_{298.15}^\circ$ (J/mol)	S_T° (J/mol K)	$-(G_T^\circ - H_{298.15}^\circ)/T$ (J/mol K)
298.15	20.786	0	175.375	175.375
300	20.786	38	175.504	175.376
350	20.786	1078	178.708	175.629
400	20.786	2117	181.484	176.191
450	20.786	3156	183.932	176.918
500	20.786	4196	186.122	177.731
550	20.786	5235	188.103	178.585
600	20.786	6274	189.912	179.455
600.612	20.786	6287	189.933	179.465
700	20.787	8353	193.116	181.183
800	20.790	10,432	195.892	182.852
900	20.801	12,511	198.341	184.440
1000	20.829	14,593	200.534	185.941
1100	20.888	16,678	202.522	187.360
1200	20.991	18,772	204.343	188.700
1300	21.155	20,878	206.029	189.969
1400	21.394	23,005	207.606	191.173
1500	21.717	25,160	209.092	192.319
1600	22.132	27,352	210.506	193.412

Table 14 (*continued*)

T (K)	C_p° (J/mol K)	$H_T^\circ - H_{298.15}^\circ$ (J/mol)	S_T° (J/mol K)	$-(G_T^\circ - H_{298.15}^\circ)/T$ (J/mol K)
1700	22.640	29,590	211.863	194.457
1800	23.239	31,883	213.174	195.461
1900	23.923	34,240	214.448	196.427
2000	24.680	36,670	215.694	197.359
2100	25.499	39,178	216.918	198.261
2200	26.365	41,772	218.126	199.137
2300	27.262	44,452	219.316	199.988
2400	28.174	47,224	220.495	200.818

$H_{298.15}^\circ - H_0^\circ = 6197.39\,J/mol.$

Table 15
Thermodynamic properties of the diatomic gas.

T (K)	C_p° (J/mol K)	$H_T^\circ - H_{298.15}^\circ$ (J/mol)	S_T° (J/mol K)	$-(G_T^\circ - H_{298.15}^\circ)/T$ (J/mol K)
298.15	37.550	0	278.148	278.148
300	37.554	69	278.380	278.149
350	37.659	1949	284.177	278.606
400	37.747	3835	289.212	279.624
450	37.825	5724	293.663	280.942
500	37.896	7617	297.652	282.417
550	37.965	9514	301.267	283.968
600	38.033	11,413	304.573	285.550
600.612	38.034	11,436	304.612	285.571
700	38.180	15,224	310.447	288.697
800	38.361	19,051	315.556	291.742
900	38.599	22,898	320.088	294.645
1000	38.912	26,773	324.170	297.396
1100	39.310	30,683	327.897	300.002
1200	39.790	34,638	331.337	302.472
1300	40.347	38,644	334.543	304.817
1400	40.968	42,709	337.556	307.049
1500	41.636	46,839	340.405	309.178
1600	42.335	51,038	343.114	311.215
1700	43.048	55,307	345.702	313.169
1800	43.759	59,647	348.183	315.045
1900	44.454	64,058	350.567	316.853
2000	45.122	68,537	352.865	318.596
2100	45.754	73,081	355.082	320.281
2200	46.344	77,686	357.224	321.912
2300	46.887	82,348	359.296	323.492
2400	47.382	87,062	361.302	324.265

$H_{298.15}^\circ - H_0^\circ = 10{,}603\,J/mol.$

Table 16
Vapour pressure.

T(K)	**Monatomic gas**			**Diatomic gas**		
	p (bar)	ΔG_T (J/mol)	ΔH_T (J/mol)	p (bar)	ΔG_T (J/mol)	ΔH_T (J/mol)
298.15	3.75×10^{-29}	162,534	195,200	1.13×10^{-46}	262,262	306,508
300	6.10×10^{-29}	162,050	195,189	2.43×10^{-46}	261,988	306,479
350	4.33×10^{-24}	156,549	194,892	9.97×10^{-39}	254,635	305,686
400	1.86×10^{-20}	151,093	194,572	4.94×10^{-33}	247,398	304,852
450	1.23×10^{-17}	145,679	194,225	1.29×10^{-28}	240,268	303,970
500	2.20×10^{-15}	140,305	193,850	4.31×10^{-25}	233,240	303,034
550	1.52×10^{-13}	134,970	193,443	3.22×10^{-22}	226,309	302,038
600	5.14×10^{-12}	129,673	192,999	7.83×10^{-20}	219,470	300,971
600.612	5.35×10^{-12}	129,609	192,993	8.33×10^{-20}	219,386	300,957
600.612	5.35×10^{-12}	129,609	188,211	8.33×10^{-20}	219,386	291,393
700	1.11×10^{-9}	119,988	187,238	3.20×10^{-16}	207,654	289,102
800	6.15×10^{-8}	110,446	186,301	1.55×10^{-13}	196,170	286,898
900	1.37×10^{-6}	101,018	185,403	1.84×10^{-11}	184,956	284,790
1000	1.62×10^{-5}	91,689	184,540	8.18×10^{-10}	173,972	282,776
1100	1.22×10^{-4}	82,444	183,708	1.79×10^{-8}	163,142	280,850
1200	6.46×10^{-4}	73,274	182,905	2.29×10^{-7}	152,570	279,013
1300	2.64×10^{-3}	64,170	182,132	1.95×10^{-6}	142,105	277,259

Table 16 (*continued*)

T(K)	**Monatomic gas**			**Diatomic gas**		
	p (bar)	ΔG_T (J/mol)	ΔH_T (J/mol)	p (bar)	ΔG_T (J/mol)	ΔH_T (J/mol)
1400	8.78×10^{-3}	55,124	181,389	1.21×10^{-5}	131,772	275,586
1500	2.48×10^{-2}	46,130	180,681	5.85×10^{-5}	121,557	273,988
1600	6.11×10^{-2}	37,182	180,008	2.30×10^{-4}	111,444	272,458
1700	0.135	28,275	179,367	7.65×10^{-4}	101,425	207,969
1800	0.273	19,405	178,756	2.21×10^{-3}	91,496	269,501
1900	0.512	10,568	178,168	5.69×10^{-3}	81,645	268,022
2000	0.899	1762	177,600	1.33×10^{-2}	71,876	266,505
2100	1.495	−7016	177,041	2.84×10^{-2}	62,183	264,915

$\Delta H_0^\circ(Pb_1) = 195{,}849\,J/mol.;\ \Delta H_0^\circ(Pb_2) = 309{,}598\,J/mol.$

Table 17
Equilibrium vapour pressure.

Total p (bar)	Pb_1 p (bar)	Pb_2 p (bar)	T (K)
10^{-15}	1.00×10^{-15}	1.26×10^{-25}	492
10^{-14}	1.00×10^{-14}	4.59×10^{-24}	517
10^{-13}	1.00×10^{-13}	1.68×10^{-22}	545
10^{-12}	1.00×10^{-12}	6.10×10^{-21}	576
10^{-11}	1.00×10^{-11}	2.19×10^{-19}	611
10^{-10}	1.00×10^{-10}	7.73×10^{-18}	651
10^{-9}	1.00×10^{-9}	2.71×10^{-16}	698
10^{-8}	1.00×10^{-8}	9.47×10^{-15}	751
10^{-7}	1.00×10^{-7}	3.29×10^{-13}	814
10^{-6}	1.00×10^{-6}	1.13×10^{-11}	889
10^{-5}	1.00×10^{-5}	3.89×10^{-10}	979
10^{-4}	1.00×10^{-4}	1.32×10^{-8}	1089
10^{-3}	1.00×10^{-3}	4.44×10^{-7}	1229
10^{-2}	9.99×10^{-3}	1.47×10^{-5}	1412
10^{-1}	9.95×10^{-2}	4.81×10^{-4}	1660
1	0.985	1.52×10^{-2}	2017.12
NBP	0.998	1.55×10^{-2}	2019.62

NBP=Normal boiling point at one atmosphere pressure (1.01325 bar).

11. Deviations of experimental values not used in the evaluation

See Tables 9 and 10.

12. Thermodynamic tables

See Tables 11–17.

Acknowledgement

The author is indebted to Malcolm Rand for calculating the thermodynamic properties of the diatomic gas

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2012.08.004.

References

[1] R.E. Bedford, G. Bonnier, H. Maas, F. Pavese, Metrologia 33 (1996) 133–154.
[2] T.B. Douglas, J.Res. Nat.Bur.Stand. 73A (1969) 451–470.
[3] R.D. Weir, R.N. Goldberg, J. Chem. Thermodynamics 28 (1996) 261–276.
[4] M.E. Wieser, T.B. Coplen, Pure Appl. Chem. 82 (2011) 359–396.

[5] R. Hultgren, P.D. Desai, D.T. Hawkins, M. Gleiser, K.K. Kelley, D.D. Wagman, Selected Values of the Thermodynamic Properties of the Elements, American Society For Metals, Metals Park, Ohio, 1973.
[6] L.V. Gurvich, I.V. Veyts, V.A. Medvedev, G.A. Khachkuruzov, V.S. Yungman, G.A. Bergman, V.F. Baybuz, V.S. Iorish, G.N. Yurkov, S.I. Gorbov, I.I. Nazarenko, O.V. Dorofeyeva, L.F. Kuratova, E.L. Osina, A.V. Gusarov, V.Ya. Leonidov, I.N. Przheval'skiy, A.L. Rogatskiy, Yu.M. Efremov, V.G. Ryabova, V.Yu. Zitserman, Yu.G. Hayt, E.A. Shenyavskaya, M.E. Efimov, V.A. Kulemza, Yu.S. Khodeyev, S.E. Tomberg, V.N. Vdovin, A.Ya. Yakobson, M.S. Demidova, in: Fourth edition,in: L.V. Gurvich, I.V. Veyts, C.B. Alcock (Eds.), Thermodynamic Properties of Individual Substances, 2, Hemisphere Publishing Corp., New York, 1991.
[7] M.W..Chase Jr., NIST-JANAF Thermochemical Tables, Fourth edition, J. Phys. Chem. Ref. Data, Monograph no. 9, 1998.
[8] J.D. Cox, D.D. Wagman, V.A. Medvedev, CODATA Key Values for Thermodynamics, Hemisphere Publishing Corp., New York, 1989.
[9] P.F. Meads, W.R. Forsythe, W.F. Giauque, J. Am. Chem. Soc. 63 (1941) 1902–1905.
[10] H.L. Bronson, A.J.C. Wilson, Can. J. Res.A 14 (1936) 181–193.
[11] A.J. Leadbetter, J. Phys. C (Proc. Phys. Soc.) Ser. 2 (1) (1968) 1481–1488.
[12] D.L. Martin, Can. J. Phys. 65 (1987) 1104–1110.
[13] N.E..Phillips, M.H..Lambert, W.R..Gardner, Unpublished work quoted in Ref. [14].
[14] N.E. Phillips, CRC Critical Rev. Solid State Sci. 2 (1972) 467–553.
[15] J.E. Neighbor, J.F. Cochran, C.A. Shiffman, Phys. Rev. 155 (1967) 384–387.
[16] M. Horowitz, A.A. ASilvidi, S.F. Malakar, J.G. Daunt, Phys. Rev. 88 (1952) 1182–1186.
[17] D.L. Decker, D.E. Mapother, R.W. Shaw, Phys. Rev. 112 (1958) 1888–1898.
[18] P.H. Keesom, B.J.C. Van Der Hoeven Jr., Phys. Lett. 3 (1963) 360–361.
[19] B.J.C. Van Der Hoeven Jr., P.H. Keesom, Phys. Rev. 127 (1965) A103–A107.
[20] L.C. Clune, B.A. Green Jr., Phys. Rev. B 1 (1970) 1459–1467.
[21] H.V. Culbert, D.E. Farrell, B.S. Chandrasekhar, Phys. Rev. B 3 (1971) 794–800.
[22] N.E. Phillips, M.H. Lambert, W.R. Gardner, Rev. Mod. Phys. 36 (1964) 131–133.
[23] D.L. Waldorf, G.A. Alers, J. Appl. Phys. 33 (1962) 3266–3289.
[24] W.H..KeesomH. Kammerlingh Onnes, Proceedings of the National Academy of Sciences, Amsterdam, Comm.Phys.Lab.Univ.Leiden, no. 143, vol. 17, 1914, p. 894.
[25] L. Malaspina, A. Gigli, G. Bardi, Ann. Chim. (Rome) 61 (1971) 482–492.
[26] S. Stølen, F. Grønvold, Thermochim Acta 327 (1999) 1–32.
[27] F. Grønvold, Priv. Comm. (1999).
[28] T.B. Douglas, J.L. Dever, J. Am. Chem. Soc. 76 (1954) 4824–4826.
[29] V.P. Onistchenko, O.F. Kutirkin, A.Yu. Bykov, High Temp.-High Press. 31 (1999) 113–118.
[30] D.R. Wood, K.L. Andrews, J. Opt. Soc. Am. 58 (1968) 818–828.
[31] H.G..Kolsky, R.M..Gilmer, F.S..Tomkins, USAEC Rept. LA-2110, 1957.
[32] P.J..Mohr, B.N..Taylor, D.B..Newell, The 2010 CODATA Recommended Values of the Fundamental Physical Constants, (Web Version 6.0), 2 June 2011, National Institute of Standards and Technology, Gaithersburg, MD 20899, USA.〈http://physics.nist.gov/constants〉.
[33] M.H. Rand, Priv. Comm. (2009).
[34] L.-E. Berg, S. Ishmial, L. Klynning, H. Martin, A. Pereira, P. Royan, Phys. Scr. 23 (1981) 1047–1051.
[35] H. Sontag, R. Weber, J. Mol. Spectrosc. 100 (1983) 75–81.
[36] M.C. Heaven, T.A. Miller, V.E. Bondybey, J. Phys. Chem. 87 (1983) 2072–2075.
[37] K.D. Setzer, J. Borkowski-Burnecki, W. Zyrnicki, E.H. Fink, J. Mol. Spectrosc. 244 (2000) 244–248.
[38] K.K. Das, H.-P. Leibermann, R.J. Buenker, G. Hirach, J. Chem. Phys. 104 (1996) 6631–6642.
[39] C.B. Alcock, P. Grieveson, J. Inst. Met. 90 (1961–1962) 304–310.
[40] H.C. Greenwood, Proc. Roy. Soc.A 82 (1909) 396–408.
[41] H.C. Greenwood, Proc. Roy. Soc.A 83 (1910) 484–490.
[42] H. von Wartenberg, Z. Elektrochem 19 (1913) 482–489.
[43] O. Ruff, B. Bergdahl, Z. Anorg. Allgem. Chem. 106 (1919) 76–94.
[44] C.K. Ingold, J. Chem. Soc. 121 (1922) 2419–2432.
[45] A.C. Egerton, Proc.Roy.Soc. A 103 (1923) 469–486.
[46] P. Harteck, Z. Physik. Chem. 134 (1928) 1–20.
[47] W. von Leitgebel, Z. Anorg. Allgem. Chem. 202 (1931) 305–324.
[48] W.,Brüll, Vortrag vor der Chemischen Gessellschaft in Göttingen, Dec 1932.
[49] J. Fischer, Z. Anorg. Allgem. Chem. 219 (1934) 1–16.
[50] E. Baur, A. Brunner, Helv. Chim. Acta 17 (1932) 958–969.
[51] A.A. Granovskaya, A.P. Lyubimov, Zh. Fiz. Khim. 27 (1953) 1437–1442.
[52] A.T. Aldred, J.N. Pratt, Trans. Faraday Soc. 57 (1961) 611–618.
[53] V.S. Esyutin, I.A. Onaev, Tr. Inst. Met. Obogashch. Akad. Nauk. Kaz. SSR. 4 (1962) 43–47.
[54] G.W..Denison, Diss. Univ.Oklahoma, 1962.
[55] P. Goldfinger, M. Jeunehomme, Trans. Faraday Soc. 59 (1963) 2851–2867.
[56] P. Sieben, G. Schmahl, Giesserei. Tech. Wiss. Beih. Giessereiw. Metallk. 18 (1966) 197–211.
[57] P.A. Rice, D.V. Ragone, J. Chem. Phys. 45 (1966) 4141–4145.
[58] R.Zh. Khobdabergenov, V.I. Nesterov, E.T. Ivragimov, A.A. Shendyapin, E.I. Vasharat'yan, E.I. Kalinin, Tr. Inst. Met. Obogashch. Akad. Nauk Kaz. SSR, Inst. Met. A.A. Baikov. 26 (1967) 37–41.
[59] H.E.J. Schins, R.W.M. van Wijk, B. Dorpema, Z. Metallkde 62 (1971) 330–336.
[60] R.J.L. Andon, J.F. Martin, K.C. Mills, J. Chem. Soc. (A) (1971) 1788–1791.
[61] A. Desideri, V. Piacente, J. Chem. Eng. Data 18 (1973) 90–93.
[62] I.I..Kopach, E.E..Lukashenko, Nov.Teor.Tekhnol.Metall.Protsessov [New achievements in theory and technology of metallurgical processes], Krasnoyarsk, Russia, 1973, pp. 63–66.
[63] O.N. Muradova, Uch. Zap. Ul'yanovsk. Gos. Ped. Inst. 27 (1974) 26–34.
[64] A. Landsberg, A. Adams, S.D. Hill, U.S. Bur, Mines Rept. Invest. 8207 (1977).
[65] D.A. Katsov, B.V. L'vov, L.K. Polzik, Yu.V. Semenov, Zh. Prikl. Specktrosc 26 (1977) 598–605;
D.A. Katsov, B.V. L'vov, L.K. Polzik, Yu.V. Semenov, J. Appl. Spectrosc. 26 (1977) 430–436.
[66] S.E. Vaisburd, L.Sh. Tsemekhman, A.V. Taberko, Yu.AS. Karasev, in: A.I. Manokhin (Ed.), Protsessy Tsvetn.Metall.Nizk Davleniiakh [The processes of non-ferrous metals at low pressures], Izd.Nauka,, Moscow, 1983, pp. 120–128.
[67] J.P. Botor, P.M. Kapias, J.G. Edwards, Z. Metallkde 78 (1987) 491–497.
[68] W.H. Rodebush, A.L. Dixon, Phys. Rev. 26 (1925) 851–858.
[69] W.H. Rodebush, A.L. Dixon, J. Am. Chem. Soc. 47 (1925) 1036–1043.
[70] A.D. Kirshenbaum, J.A. Cahill, Trans. ASM. 55 (1962) 844–847.
[71] A.D. Kirshenbaum, J.A. Cahill, J. Inorg. Nucl.Chem. 25 (1963) 232–234.
[72] J.H. Kim, A. Cosgarea Jr., J. Chem. Phys. 44 (1966) 806–809.
[73] J. Bodhansky, H.E.J. Schins, J. Appl. Phys. 36 (1965) 3683–3684.
[74] J. Bodhansky, H.E.J. Schins, J. Phys. Chem. 71 (1967) 215–217.
[75] D.T. Hawkins, R. Hultgren, Trans. Met. Soc. AIME 239 (1967) 1046–1049.
[76] Y.A. Chang, G.C. Williams, M. Lathrup, I. Gyuk, Acta Met. 19 (1971) 795–805.
[77] D.H. Shiu, Z.A. Munir, Met. Trans. 2 (1971) 2953–2955.
[78] R. Prasad, V. Venogopal, D.D. Sood, J. Chem. Thermodynamics 9 (1977) 765–772.
[79] A. Zajaczkowski, J. Botor, Bull. Electrochem. 7 (1) (1991) 378–383.
[80] R.E. Honig, J. Chem. Phys. 21 (1953) 573–574.
[81] J. Drowart, R.E. Honig, J. Phys. Chem. 61 (1957) 980–985.
[82] K.A. Gingerich, D.L. Cooke, F. Miller, J. Chem. Phys. 64 (1976) 4027–4033.
[83] J.-P. Nabot, C. Chatillon, Z. Metallkde. 81 (1990) 100–104.
[84] W.H. Keesom, D.H. Andrews, Comm. Phys. Lab.Univ. Leiden No.185a, 1927 and Proc. Acad. Sci. Amsterdam 30 (1927) 434–443.
[85] W.H. Keesom, J.N. van den Ende, Comm. Phys. Lab. Univ. Leiden No.203d, 1930 and Proc. Acad. Sci. Amsterdam 33 (1930) 243–254;
W.H. Keesom, J.N. van den Ende, Comm. Phys. Lab. Univ. Leiden No.203d, 1930 and Proc. Acad. Sci. Amsterdam 34 (1931) 210–211.
[86] J. Bonnerot, Rev. Phys. Appliquée 1 (1966) 61–67.
[87] C. Drucker, Z. Physik. Chem. 130 (1927) 673–690.
[88] A. Eucken, F. Schwers, Ber. Deut. Physik. Ges. 15 (1913) 578–588.
[89] W. Nernst, Ann. Phys. 36 (1911) 395–439.
[90] H. Barschall, Z. Elektrochem. 17 (1911) 341.
[91] E.H. Griffiths, E. Griffiths, Phil.Trans.R. Soc. A 213 (1913) 119–185;
E.H. Griffiths, E. Griffiths, Phil. Trans. R. Soc. A 214 (1914) 319–357;
E.H. Griffiths, E. Griffiths, Proc. R. Soc. A 88 (1913) 549–560.
[92] H. Klinkhardt, Ann. Physik. 84 (1927) 167–200.
[93] R. Brandt, O. Schulz, J. Nucl. Mater. 152 (1988) 178–181.
[94] G.M. Bartenev, Zh. Tekhn. Fiz. 17 (1947) 1321–1324.
[95] G.M. Bartenev, Uch. Zap. Mosk. Gos. Univ. M.V. Lomonsova No.134, Fiz. (5) (1949) 113–133.
[96] T.E. Pochapsky, Acta Met. 1 (1953) 747–751.
[97] V.A. Kogan, V.E. Lyusternik, Inzh. Fiz. Zh. Akad Nauk Belorussk.SSR. 4 (4) (1961) 105–108.
[98] G. Cordoba, C.R. Brooks, Phys. Stat. Sol. (a) 7 (1971) 503–508;
G. Cordoba, C.R. Brooks, Phys. Stat. Sol. (a) 11 (1971) 749–754.
[99] A.L. Dixon, W.H. Rodebush, J. Am. Chem. Soc. 49 (1927) 1162–1174.
[100] S.N. Banchilla, L.P. Filippov, Teplofiz. Vys. Temp. 11 (1973) 668–671;
S.N. Banchilla, L.P. Filippov, High Temp. 11 (1973) 602–605.
[101] N. Stücker, Sitzb. König. Akad. Wiss. Wein. 114 (1905) 657.
[102] A. Magnus, Ann. Phys. (Leipzig) 31 (1910) 597–608.
[103] P. Schübel, Z. Anorg, Allgem. Chem. 87 (1914) 81–119.
[104] F. Wüst, A. Meuthen, A. Durrer, Forsch. Gebiete. Ingenieurw. 204 (1918).
[105] I. Iitaka, Sci. Rept. Tohoku. Univ. 8 (1919) 99–113.
[106] S. Umino, Sci. Rept. Tohoku Univ. 15 (1926) 597–617.
[107] J.H. Awbery, E. Griffiths, Proc. Phys. Soc. 38 (1926) 378–398.
[108] P.H..Sommelet, USAEC Rept. UCRL-16303, 1965.
[109] F. Gläser, Metallurgie. 1 (1904) 121–128.
[110] H. Schimpff, Z. Phys. Chem. 71 (1910) 257–300.
[111] G. Mishra, R. Kumar, Trans. Indian Inst. Metals 20 (1967) 25–28.
[112] G. Pottlacher, H. Jäger, Int. J. Thermophys. 11 (1990) 719–729.
[113] W. Kramer, J. Nölting, Acta Met. 20 (1972) 1353–1359.